Plan — R10.2 Structured-Option Torque-Path TD3 (Stage 2, corrected coordinate)

One structured θ at READY $\to$ a scaffold-relative desired-torque path $\to$ a deterministic slew tracker. Exploration is temporally coherent *by construction*; zero-θ is the medoid scaffold bit-exact.

HyMeKo coin-delivery campaign (`recovery/coin-r9-causal-residual-delivery`, parent `4d983d4f`)

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Abstract

Stage 1B proved a phase-shaped moving-precursor capture delivers strict K6 from `HOME_V1` (3/3 seeds, frozen downstream). The Stage-2 audit (`4d983d4f`) then established a *negative-first* result: the previous action coordinate — a per-step slew-normalised residual $u = \mathrm{clip}(u_{\pi_0}(\phi) + \alpha \tanh \delta_\theta(s), \pm 1)$ whose δ_θ is re-evaluated *each step* — makes vanilla TD3 explore with per-step IID Gaussian noise that *integrates into the actuator preload* ($\delta\tau_T \approx \alpha \Delta\tau_{\max} \sum_t \varepsilon_t$), removing the delivery at every tested scale, while a temporally-*coherent* correction recovers 8/8. The fix is not to abandon TD3 but to **move the action out of the per-step slew-increment coordinate**. The learned decision is made *once, at READY*, as a bounded structured option $\theta = (\Delta s, \Delta p, \Delta b,\ c_1, c_2 \in \mathbb{R}^4,\ \Delta\tau_T \in \mathbb{R}^4) \in \mathbb{R}^{15}$; the per-step physical action is a *deterministic tracker* of the resulting smooth desired-torque path $\tau_{\mathrm{des}}(\phi) = \tau_0^\theta(\phi) + \sum_k c_k \psi_k(\phi) + \psi_T(\phi)\,\Delta\tau_T$, so exploration is temporally coherent by construction. **This is a coordinate-freeze plan, not yet a learning claim.** Three identity gates are verified *before any training* (zero-$\theta \equiv$ medoid scaffold bit-exact on the *same* code path $\Rightarrow$ strict K6); then episodic-only exploration admissibility (incl. terminal-offset tracking / no preload random-walk); only then a bounded 3-seed TD3 whose success is *improvement over the frozen scaffold under perturbation*, never nominal K6 (the scaffold already has it). No SAC/PPO until this coordinate is frozen.

1 Scope and goal

Goal. Freeze a *correct* structured-option torque-path action coordinate over the frozen Stage-1B scaffold, prove its zero-θ identity and exploration admissibility, and run one bounded reward-driven TD3 campaign whose claim is *state-dependent robustness improvement over the frozen scaffold*, deployed with no CEM/teacher/state-edit in the loop. **Deployed chain (unchanged physically):**

*HOME_V1 $\to$ frozen analytic transit $\to$ READY $\to$ **structured-option torque-path capture** $\to$ frozen APPROACH $\to$ HANDOFF_RESET $\to$ frozen R2 $\to$ coast/settle $\to$ strict K6.*

Non-goals (explicit stops). No SAC/PPO (deferred to R10.4, only after this coordinate freezes); no 24-seed reproduction (R10.3); no variable-geometry / variable-coin work (R11+); no Hamiltonian/energy experiments (R15). No re-opening the retired per-step IID / cradle-centred-$\tau^\star$ path. This is the bounded coordinate-freeze + 3-seed smoke only.

2 What changed from the retired coordinate (`4d983d4f` audit)

- **Decision cadence.** Retired: a residual $\delta_\theta(s)$ re-evaluated *every step* (per-step action). New: one θ emitted *once at READY* (episode-fixed); the per-step action is a fixed deterministic function of θ and the phase.

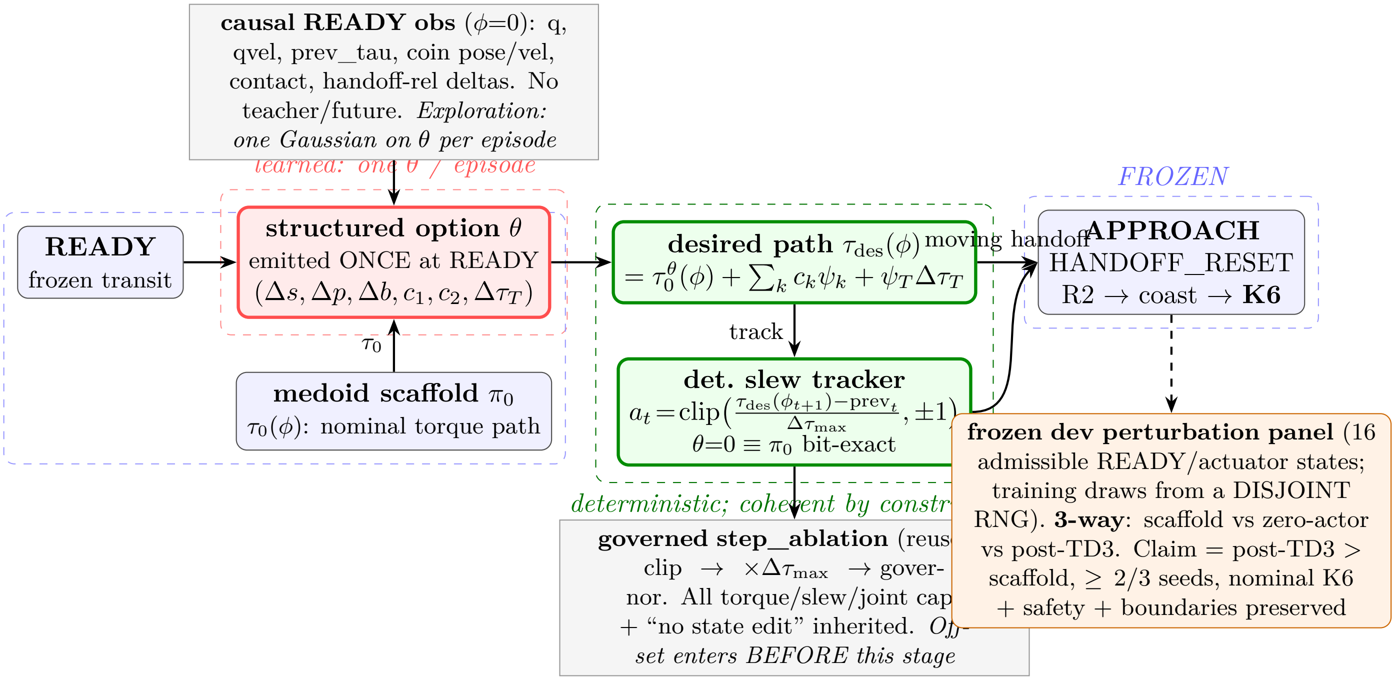

Figure 1: The learned decision is a single structured option θ emitted at READY (red); a deterministic tracker (green) converts the resulting smooth desired-torque path into per-step slew-normalised commands through the *same* governed `step_ablation` stack the scaffold uses. At θ=0 the tracker is the medoid scaffold π_0 (blue) bit-exact. The claim is measured on the frozen dev perturbation panel against scaffold and zero-actor baselines.

- **Exploration geometry.** Retired: `train_semi_mdp`'s Gaussian is added to the per-step residual ⇒ IID slew-increment noise that random-walks the preload. New: the *same* engine's per-option Gaussian is added to θ ⇒ one coherent structured perturbation per episode. *The coordinate change is what makes the reused engine admissible.*
- **Preload.** Retired: terminal preload is a hidden noise integral. New: preload change is the explicit, bounded terminal-offset block $\Delta\tau_T$ with a fixed basis $\psi_T(0)$=0, $\psi_T(1)$=1; requested-vs-executed offset is gated (`TERMINAL_OFFSET_TRACKING` / `NO_PRELOAD_RANDOM_WALK`).
- **Reward reference.** Retired: cradle-centred $\tau^\star$ preload metric — *invalidated* (medoid terminal $\|\text{prev} - \tau^\star\| \approx 0.88$, not 0). New: shaping is *scaffold-phase-tube*-relative only $(q - q_0(\phi),\ \dot{q} - \dot{q}_0(\phi),\ \text{prev} - \tau_0(\phi))$; every $\tau^\star$ term removed.

3 The action coordinate (interface + contracts)

Let π_0 be the frozen medoid scaffold (structural-parameter medoid of the 3 Stage-1B K6 solutions; `freeze_scaffold`), with nominal params $(s, p, b, \text{residual})$ and per-step slew-normalised action $a_{\pi_0}(\phi) = \text{clip}(\text{target}(\phi) - \text{prev}, -\Delta\tau_{\max}, \Delta\tau_{\max})/\Delta\tau_{\max} \in [-1,1]^4$ where $\Delta\tau_{\max} = \tau_{\text{rate}}$control_dt (`PhaseShapeCapture.scaffold`

Structured option (emitted once at READY): $\theta = (\Delta s, \Delta p, \Delta b, c_1, c_2, \Delta\tau_T) \in \mathbb{R}^{15}$, decoded from the actor's tanh-bounded output $\hat{z} \in [-1,1]^{15}$ by fixed, pre-registered per-block scales (a `StructuredOption` dataclass; $\hat{z}$=0 ⇒ all offsets exactly 0).

Structurally-reshaped nominal. $\tau_0^{\theta}(\phi)$ is the scaffold nominal under the perturbed params $(s{+}\Delta s,\ p{+}\Delta p,\ b{+}\Delta b)$; at θ=0 these are bit-identical to (s, p, b) (since $x + 0.0 = x$), so the quintic coefficients and preload blend are bit-identical.

Additive torque-path offset. $o_\theta(\phi) = \sum_{k\in\{1,2\}} c_k\, \psi_k(\phi) + \psi_T(\phi)\, \Delta\tau_T$ with transient bases

$\psi_k(0)=\psi_k(1)=0$ (a knot spline pinned to 0 at both endpoints — modifies the middle, no unnoticed endpoint drift) and terminal basis $\psi_T(0)=0, \psi_T(1)=1$ (a monotone ramp; the preload change is explicit and bounded, never a noise integral).

Deterministic per-step tracker (the key). The desired torque path is $\tau_{\text{des}}(\phi) = \tau_0^\theta(\phi) + o_\theta(\phi)$, tracked by

$$a_t = \operatorname{clip}\Big(\tfrac{\tau_{\text{des}}(\phi_{t+1})-\text{prev}_t}{\Delta\tau_{\max}}, -1, 1\Big), \qquad \text{prev}_{t+1} = \operatorname{clip}(\text{prev}_t + \operatorname{clip}(a_t, \pm 1)\,\Delta\tau_{\max},\ \text{lo}, \text{hi}),$$

a tracking form that cannot accumulate action error (it re-references the *current* $\texttt{prev}_t$ each step). No per-step residual noise is ever added after this conversion.

Bit-exact realisation (float-exactness contract). A literal $(\tau_{\text{des}} - \text{prev})/\Delta\tau_{\max}$ would route a_{π_0} through a $\Delta\tau_{\max}$-divide-then-multiply round-trip and lose bit-exactness at $\theta=0$. Instead the tracker is computed as $a_t = \operatorname{clip}\big(a^\theta_{\pi_0}(\phi) + o_\theta(\phi)/\Delta\tau_{\max}, -1, 1\big)$, injecting the offset into the scaffold's *unnormalised* numerator $\operatorname{clip}(\text{target} - \text{prev}, -\Delta\tau_{\max}, \Delta\tau_{\max})$ via the reused `scaffold_action/apply_step`. This is algebraically the same τ_{des} tracker with $\tau_0^\theta(\phi) = \text{prev} + \operatorname{clip}(\text{target} - \text{prev}, -\Delta\tau_{\max}, \Delta\tau_{\max})$, and at $\theta=0$ ($o_\theta \equiv 0$) reduces to $\operatorname{clip}(a_{\pi_0}, -1, 1) = a_{\pi_0}$ *on the identical code path.*

Contracts inherited from the scaffold (§CLAUDE.md 2 risk): every command passes `step_ablation` + governor + the final `clip`$\to \times\Delta\tau_{\max}$; torque / slew / joint-speed caps and "no state edit, no hidden force" hold by *reuse*, not re-implementation. The one new degree of freedom is the explicit terminal preload $\Delta\tau_T$, which the tracking-gate keeps inside the scaffold's tight-on-preload deliverable band.

4 The frozen contract

- Frozen and imported as-is: `HOME_STATE_V1`, the analytic HOME→READY transit (`eb09a16a`), the Stage-1B scaffold (`moving_precapture`, `d6974b95: PhaseShapeCapture`, `scaffold_action`, `apply_step`, `FrozenDownstream`, `HandoffReference`), APPROACH / HANDOFF_RESET / R2 / coast, and physics / safety / strict-K6. `git diff` empty on all.
- The structured option controls *only* READY→capture; the frozen downstream generates the rest. No teacher / future info; no state edit; no online CEM once training begins.
- `s4`/`s7` untouched (validation-only, *not read* here); `f1--f4 SEALED`; development perturbations only, drawn from a training RNG *disjoint* from the frozen dev panel. No tag moved.

5 Affected files

CORE.YAML items touched: none (empty list; escalation not required). **Reuse (read-only):** `option_rl.core` (`OptionEnv`, `OptionTransition`, `smdp_target`), `option_rl.agents` (`train_semi_mdp`, `make_actor`, `SemiMDPConfig` — the dim-generic TD3/SAC engine); `moving_precapture` (`PhaseShapeCapture`, `scaffold_action`, `apply_step`, `FrozenDownstream`); `capture_rl` (`freeze_scaffold`, `capture_observation`, `perturbation_panel`, `perturb_ready`, `evaluate_on_panel`, `make_zero_actor`); `kinetic_authority_unlock` (`zero_init_detector`).
New:

- `coin_delivery/theta_option/torque_path_option.py` — the action module: `StructuredOption` + θ-decoder, the phase bases ψ_k, ψ_T, `TorquePathCaptureRoll` (reuses `scaffold_action` / `apply_step`; injects $o_\theta/\Delta\tau_{\max}$), the nominal phase-tube recorder `record_phase_tube`, and the terminal-offset audit `terminal_offset_tracking`. Pure action coordinate; no reward, no trainer.
- `coin_delivery/theta_option/torque_path_env.py` — `StructuredOptionCaptureEnv` (an `OptionEnv`: reset → perturbed-READY causal obs; step(θ) → torque-path roll + frozen downstream + terminal reward), the phase-tube shaping reward `phase_tube_reward`, and the training-perturbation sampler (disjoint from the dev panel).

- `experiments/coin_kinetic_structured_option_gates.py` — the pre-training gate verifier: the three identity gates → terminal-offset tracking → $\leq$ 3-scale exploration admissibility. Imports no trainer; emits `reports/2026-07-28-r10-structured-option-torque-path-td3/gates.json`.
- `experiments/coin_kinetic_structured_option_td3.py` — the campaign (only after gates **PASS**): 3-seed TD3, $\leq$ 400 option-episodes/seed, eval every 25 (no exploration), 3-way frozen-panel compare, gate verdict, provenance.
- `tests/test_torque_path_option.py`; `reports/2026-07-28-r10-structured-option-torque-path-td3.md`.
- *Additive, non-core:* `capture_rl.make_zero_actor` gains an optional `act_dim` (default 4; preserves the existing test) so a 15-D zero-θ actor is built by reuse, not a near-copy (§6.1).

6 Interface changes

- `decode_theta(`$\hat{z} \in [-1,1]^{15}$`, scales) → StructuredOption`. **Pre:** $\|\hat{z}\|_\infty \leq 1$. **Post:** deterministic; $\hat{z}$=0 ⇒ every offset exactly 0.0 (identity precondition).
- `TorquePathCaptureRoll(ready, ref, stack, pi0, coin0, scales).rollout(`$\hat{z}$`) → {snapshot, obs, acts, prev, tube_dev}`. **Post:** $\hat{z}$=0 reproduces `PhaseShapeCapture.roll(pi0)` *bit-exact* (asserted by the identity gate); every step passes the governed stack.
- `record_phase_tube(ready, ref, stack, pi0) → {q0(`ϕ`), qvel0(`ϕ`), tau0(`ϕ`)}` — the medoid reference trajectory; the *only* shaping reference (no $\tau^\star$).
- `StructuredOptionCaptureEnv` — `reset(idx)→s` (perturbed-READY causal obs at ϕ=0); `step(`$\hat{z}$`)→(s2,R,done,info` rolls the torque-path capture + frozen downstream, returns the strict-K6-dominated terminal reward; `info` carries `tau`, k6, safe, min_dtz, boundary counts, terminal-offset error, tube deviation. Each option is terminal (γ^τ bootstrap collapses to the reward).
- `evaluate_on_panel(policy, …)` (reused) — K6 rate + Wilson CI + safety + one-reset frac on the frozen dev panel; identical for scaffold / zero-actor / post-TD3.

7 Reward (K6-dominant; shaping is phase-tube-relative only)

$R = w_{K6}\,\mathbf{1}[\text{strict K6}] + w_d\,\rho(\text{min\_dtz}) - w_s\,\mathbf{1}[\neg\text{safe}] - w_b\,(\text{boundary violations}) - w_\phi\,\bar{S}_\theta$, where the *only* shaping term is the bounded phase-tube deviation $\bar{S}_\theta = \frac{1}{N}\sum_t \big(\lambda_q\|q - q_0(\phi)\|^2 + \lambda_v\|\dot{q} - \dot{q}_0(\phi)\|^2 + \lambda_\tau\|\text{prev} - \tau_0(\phi)\|^2\big)$. Weights are set so $w_{K6} \gg \sup|\text{shaping}|$: strict K6 cannot be out-farmed by tube-hugging, and the tube is shaping, never a target (it *is* the zero-θ trajectory, so hugging it → the scaffold, not an exploit). **No cradle-$\tau^\star$ term anywhere.**

8 Test strategy

Unit: θ-decoder ($\hat{z}$=0 ⇒ exact-0 offsets; scale bounds; determinism); phase bases ($\psi_k(0)$=$\psi_k(1)$=0, $\psi_T(0)$=0, $\psi_T(1)$=1); phase-tube recorder shape/finiteness; reward K6-dominance (shaping cannot flip the sign of a K6 vs non-K6 pair); training-perturbation sampler disjoint from the dev panel. **Integration / identity (load-bearing, before training):** `TORQUE_PATH_ZERO_DELTA_IDENTITY` (real 15-D zero-init actor ⇒ θ=0 ⇒ per-step action bit-exact a_{π_0}), `TORQUE_PATH_BIT_EXACT_SCAFFOLD` (terminal qpos/qvel/prev bit-exact to `PhaseShapeCapture.roll(pi0)`), `OPTION_ZERO_POLICY_K6` (zero-θ → frozen downstream → strict K6). **Terminal-offset tracking:** requested $\Delta\tau_T$ vs executed terminal prev $-\ \tau_0(1)$ agree within a slew/contact tolerance (`NO_PRELOAD_RANDOM_WALK`). **Smoke (production scale, 1 seed):** a short TD3 run over the real dev-scale env (real mujoco, full downstream horizon) executes end-to-end and improves panel K6 over the zero-actor, else **HALT** and localise (reward, obs, exploration scale). **Boundary:** exactly-one HANDOFF_RESET; deterministic replay given the seed. **Coverage:** every new function/method exercised; the identity gate is the regression test that would fail against a non-bit-exact tracker.

9 Performance budget

One option (capture ∼20 steps + downstream horizon 80) ≈ 0.1–0.2 s. Gates ($\leq$ 3 scales × a small panel) ≈ 1–3 min. TD3: $\leq$ 400 options/seed, eval every 25, 3 seeds ≈ 15–35 min, *checkpointed per seed*. **Peak RSS** < 2 **GB** (a 31→256→256→15 actor + two critics; hard cap 16 GB, `systemd-run` not required at this size but RSS reported on exit). Unit suite target < 180 s. Wall estimate reconciled against the Stage-1B/audit baselines before launch (§11 > 2× rule); measured median/IQR/worst over $\geq$ 5 eval repeats for any reported latency.

10 Risk anticipation

Performance-contract preservation. The new torque-path branch inherits *every* cap the scaffold enforced because it *reuses* `apply_step` (the identical clip → $\times\Delta\tau_{\max}$ → governor → `step_ablation` stage); the offset enters *before* that stage and is itself bounded ($c_k, \Delta\tau_T$ from a tanh-bounded actor × fixed scales). Zero-θ is bit-exact π_0, so the worst case is never worse than the proven scaffold. **The float-exactness trap.** A literal τ_{des}-tracker loses the θ=0 identity to a $/\Delta\tau_{\max}\cdot\Delta\tau_{\max}$ round-trip; the numerator-injection realisation avoids it and is the reason the identity gate is bit-exact rather than "≈". **The scientific trap.** Nominal K6 is necessary-not-sufficient (π_0 has it); the claim rides entirely on paired panel improvement over the frozen scaffold in $\geq 2/3$ seeds, on a panel fixed *before* any TD3 run and never tuned to results. **Preload random-walk (the retired failure).** Structurally prevented: exploration is episode-fixed on θ, and $\Delta\tau_T$ has an explicit fixed basis — the `TERMINAL_OFFSET_TRACKING` gate proves the executed preload matches the requested one under clipping/slew/contact. **Worst-case input.** Production scale here = the frozen dev panel (16 admissible READY / actuator-state perturbations, `sig_q`=0.02, `sig_v`=0.06, `sig_tau`=0.05) through the full downstream horizon 80 under real mujoco 3.10, × 3 seeds × 400 options — not a toy fixture; the 1-seed production smoke exercises exactly this before the multi-seed queue.

11 Rollback path

New files only, plus one additive `act_dim` kwarg on `make_zero_actor`; rollback = delete the new files and revert the kwarg. The zero-θ bit-exact identity guarantees a revert to π_0 (`d6974b95`) with no behaviour change. Each checkpoint (`update0`/`best_val`/`final`) is saved per seed; a regressing post-TD3 reverts to the zero-actor (≡ scaffold).

12 Gate ladder + stop logic

1. **(before training)** `TORQUE_PATH_ZERO_DELTA_IDENTITY_PASS`, `TORQUE_PATH_BIT_EXACT_SCAFFOLD_PASS`, `OPTION_ZERO_POLICY_K6_PASS`. Any failure ⇒ **HALT** (the coordinate is wrong; do not train).
2. **(before training)** `TERMINAL_OFFSET_TRACKING_PASS` + `STRUCTURED_THETA_EXPLORATION_ADMISSIBLE`: the smallest of $\leq$ 3 pre-registered episodic scales giving *both* preserved nominal successes and safe near-boundary failures and genuinely distinct trajectories, with executed $\Delta\tau_T$ matching the request (`NO_PRELOAD_RANDOM_WALK`). If no scale is admissible ⇒ **HALT** and re-scope the scales (not the coordinate).
3. **(training)** `STRUCTURED_OPTION_TD3_IMPROVES_OVER_SCAFFOLD_PASS`: post-TD3 beats the frozen scaffold (paired, same panel members) in $\geq 2/3$ seeds while *preserving* nominal HOME_V1 K6, safety, and every boundary contract. Else emit only `STRUCTURED_OPTION_TD3_NO_IMPROVEMENT_WITHIN_FROZEN_BUDGET` — *never* an RL-impossibility claim.

Stop logic. Each numbered gate is a *boundary*: verify, report, and stop for review before the next. After the TD3 verdict, commit and **HALT** — no 24-seed reproduction (R10.3), no SAC/PPO (R10.4), no energy experiments. If a gate fails, **HALT** and localise within the current coordinate; do not widen scope.

13 Empty-plan-dir hygiene

If abandoned before the action module lands, delete `docs/plans/2026-07-28-r10-structured-option-torque-path-td3` this session. The retired `docs/plans/2026-07-28-td3-capture-reward-driven/` is *kept* as the audited-negative record of the superseded coordinate (its `plan.pdf` documents why the per-step residual was refuted).